# Md Sakib Hasan Emon

**Computer Science & Engineering Graduate · AI / ML / NLP Researcher**

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## SUMMARY

Computer Science and Engineering graduate with a strong academic focus on Machine Learning, Data Mining, Deep Learning, and Natural Language Processing. Experienced in data collection, preprocessing, feature engineering, model training, and evaluation using Python-based ML frameworks. Actively engaged in research on mental health prediction from Bangla–English code-mixed (Banglish) social media data. Seeking a fully funded Master's programme in Artificial Intelligence, Machine Learning, or Data Science to advance research in Natural Language Processing and Explainable AI.

## EDUCATION

### B.Sc. in Computer Science & Engineering
2022 – May 2026

Green University of Bangladesh · CGPA 3.47 / 4.00

### Higher Secondary Certificate (HSC) — Science
2018 – 2020

Mymensingh Govt. College · GPA 5.00 / 5.00

### Secondary School Certificate (SSC) — Science
2018

Chattogram Cantonment Board High School · GPA 5.00 / 5.00

## RESEARCH EXPERIENCE

### Bachelor's Thesis — Banglish Mental Health Classification using Transformer-Based Multi-Task Learning
2025 – Present

Department of Computer Science and Engineering, Green University of Bangladesh

- Constructed and annotated a Banglish mental health dataset of 56,500+ samples across six categories and three severity levels.
- Developed a Transformer-based multi-task learning framework (BanglaBERT-MTL-AP) with Attention Pooling for joint classification and severity prediction.
- Applied Explainable AI (Grad×Input) for model interpretability and transparent decision support.
- Achieved 91.4% classification accuracy (Macro-F1 0.918) and 92.2% severity accuracy (Macro-F1 0.926).

## PUBLICATIONS

### Mental Health Classification from Social Media Data in Banglish using Transformer-Based Multi-Task Learning and Explainable AI

Submitted to the IEEE CSDE 2026 Conference (Under Review)

## WORK EXPERIENCE

### Software QA Intern — QA Harbour
2025

- Executed structured manual and functional test cases across web-based applications.
- Documented test scenarios, identified and reported defects, and supported regression testing cycles.

### Software Testing & QA Contributor — UNITY AID (IDP)
2025

Green University of Bangladesh

- Designed and executed manual test cases; performed functional and usability testing across all major user flows.
- Verified role-based access control and form validation, improving platform reliability.

## SELECTED PROJECTS

### Banglish Mental Health Text Classifier — Web App

Python · FastAPI · Explainable NLP · github.com/EMON-26/banglish-mental-health-classifier

- Companion tool to the thesis: a transparent, lexicon-based engine that weak-labeled the 56,500-sample training dataset.
- Built a FastAPI backend with live single-text and batch-CSV classification, plus phrase-level explainability.

### UNITY AID — Community Emergency Response Platform

Software Engineering · Web Platform · QA · github.com/EMON-26/unity-aid-testing

- Web-based emergency coordination platform with role-based access control and location-aware workflows.
- Led QA through systematic test case design, defect reporting, and validation documentation.

## TECHNICAL SKILLS

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|---|---|---|---|
| **Programming** | Python, Java, C | **Tools** | Git, GitHub, Linux, Bash, LaTeX, Streamlit |
| **Machine Learning** | Scikit-learn, PyTorch, TensorFlow, Transformers | **Backend / Web** | FastAPI, REST APIs |
| **Data Science** | Pandas, NumPy, Matplotlib, NLTK, SHAP | **Languages** | Bangla (Native), English (B2) |

## AWARDS & EXTRACURRICULAR

- Dean's Award — Green University of Bangladesh (Final Semester, 2026)
- BNCC Annual & Basic Training Encampment Certificates; Science Fair Participation Certificate
- Recognition for contributions to the UNITY AID Integrated Design Project
- Former Member, Green University Debate Club · Former Cadet, Bangladesh National Cadet Corps (BNCC)

## REFERENCES

**Dr. Md. Solaiman Mia** — Assistant Professor, Dept. of CSE, Green University of Bangladesh
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